Multiple Choice:

1. State which of the following is false. The capacitance of a capacitor

   A. is proportional to the cross-sectional area of the plates
   B. is proportional to the distance between the plates
   C. depends on the number of plates
   D. is proportional to the relative permittivity of the dielectric

2. A resistor with a power rating of 25 W is most likely a

   A. carbon-composition resistor.
   B. metal-film resistor.
   C. surface-mount resistor.
   D. wire-wound resistor.

3. A significant advantage, in some situations, of a toroidal coil over a solenoid is:

   A. The toroid is easier to wind
   B. The solenoid cannot carry as much current
   C. The toroid is easier to tune
   D. The magnetic flux in a toroid is practically all within the core

4. Find the e.m.f. induced in a coil of 200 turns when there is a change of flux of 30 mWb linking with it in 40 ms.

   A. -150 V C. -100 V
   B. 150 V D. 100 V

5. The charge on the plates of a capacitor is 6 mC when the potential between them is 2.4 kV. Determine the capacitance of the capacitor.

   A. 2.5 μF C. 2.0 μF
   B. 1.0 μF D. 1.5 μF

6. What formula would calculate the total inductance of inductors in series?

   A. $L_T = L_1 / L_2$
   B. $L_T = L_1 + L_2$
   C. $L_T = 1 / L_1 + L_2$
   D. $L_T = 1 / L_1 \times L_2$

7. Some wire of length 5 m and cross-sectional area 2 mm^2 has a resistance of 0.08 Ω. If the wire is drawn out until its cross-sectional area is 1 mm^2, determine the resistance of the wire.

   A. 0.32 Ω C. 0.21 Ω
   B. 0.032 Ω D. 0.021 Ω

8. What is the ability to store energy in a magnetic field called?

   A. Admittance
   B. Capacitance
   C. Resistance
   D. Inductance

9. How would you calculate the total capacitance of three capacitors in parallel?

   A. $C_T = C_1 + C_2 / C_1 - C_2 + C_3$.
   B. $C_T = C_1 + C_2 + C_3$.
   C. $C_T = C_1 + C_2 / C_1 \times C_2 + C_3$.
   D. $C_T = 1 / C_1 + 1 / C_2 + 1 / C_3$.

10. What is the ability to store energy in an electric field called?

    A. Inductance
    B. Resistance
    C. Tolerance
    D. Capacitance

11. With permeability tuning, moving the core further into a solenoidal coil:

    A. Increases the inductance
    B. Reduces the inductance
    C. has no effect on the inductance, but increases the current-carrying capacity of the coil
    D. Raises the frequency

12. Which of the following statement is false?

    A. An air capacitor is normally a variable type
    B. A paper capacitor generally has a shorter service life than most other types of capacitor
    C. An electrolytic capacitor must be used only on a.c. supplies
    D. Plastic capacitors generally operate satisfactorily under conditions of high temperature

13. A capacitor is constructed with parallel plates and has a value of 50 pF. What would be the capacitance of the capacitor if the plate area is doubled and the plate spacing is halved?

    A. 200 pF C. 100 pF
    B. 50 pF D. 75 pF

14. What electrical parameter is controlled by a potentiometer?

    A. Inductance
    B. Resistance
    C. Capacitance
    D. Field strength

15. What electrical component usually is constructed as a coil of wire?

    A. Switch C. Capacitor
    B. Diode D. Inductor

16. One precaution to observe when checking resistors with an ohmmeter is to

    A. check high resistance on the lowest ohms range
    B. check low resistance on the highest ohms range.
    C. disconnect all parallel paths.
    D. make sure your fingers are touching each test lead.

17. What is a relay?

    A. An electrically-controlled switch
    B. A current controlled amplifier
    C. An optical sensor
    D. A pass transistor

18. For how long must a charging current of 2 A be fed to a 5 μF capacitor to raise the p.d. between its plates by 500 V.

    A. 0.5 mS
    B. 1.0 mS
    C. 1.15 mS
    D. 1.25 mS

19. Carbon-composition resistors:

A. Can handle lots of power
B. Have capacitance or inductance along with resistance
C. Are comparatively nonreactive
D. Work better for ac than for dc

20. A capacitor is to be constructed so that its capacitance is 0.2 μF and to take a p.d. of 1.25 kV across its terminals. The dielectric is to be mica which, after allowing a safety factor of 2, has a dielectric strength of 50 MV/m. Find the thickness of the mica needed and the area of a plate assuming a two-plate construction. (Assume ε_r for mica to be 6).

A. 0.025 mm; 941.6 cm^2
B. 0.015 mm; 671.7 cm^2
C. 0.019 mm; 893.9 cm^2
D. 0.035 mm; 561.4 cm^2

21. The total resistance in a parallel circuit:

A. is always less than the smallest resistance
B. depends upon the voltage drop across each branch
C. could be equal to the resistance of one branch
D. depends upon the applied voltage

22. A potentiometer is a

A. three-terminal device used to vary the voltage in a circuit.
B. two-terminal device used to vary the current in a circuit.
C. fixed resistor.
D. tow-terminal device used to vary the voltage in a circuit.

23. The material separating the plates of a capacitor is the:

A. dielectric
B. semiconductor
C. resistor
D. lamination

24. An e.m.f. of 25 V is induced in a coil of 300 turns when the flux linking with it changes by 12 mWb. Find the time, in milliseconds, in which the flux makes the change.

A. 144 mS C. 114 mS
B. 120 mS D. 155 mS

25. Which of the following is combined with an inductor to make a tuned circuit?

A. Zener diode C. Resistor
B. Potentiometer D. Capacitor

26. The total capacitance of two or more capacitors in series is:

A. always less than that of the smallest capacitor
B. always greater than that of the largest capacitor
C. found by adding each of the capacitances together
D. found by adding the capacitances together and dividing by their total number

27. A potentiometer is used to:

A. Compare voltages
B. Measure power factor
C. Compare currents
D. Measure phase sequence

28. Determine the voltage across a 1000 pF capacitor to charge it with 2 μC.

A. 1 kV C. 2 kV
B. 2.5 kV D. 0.5 kV

29. The total resistance of two resistors R_1 and R_2 when connected in parallel is given by:

A. $R_1 + R_2$
B. $1/R_1 + 1/R_2$
C. $(R_1 + R_2)/R_1R_2$
D. $R_1R_2/(R_1 + R_2)$

30. Which of the following statements is false? The inductance of an inductor increases:

A. with a short, thick coil
B. when wound on an iron core
C. as the number of turns increases
D. as the cross-sectional area of the coil decreases

31. The energy stored in the magnetic field of an inductor is 80 J when the current flowing in the inductor is 2 A. Calculate the inductance of the coil.

A. 40 H C. 30 H
B. 20 H D. 50 H

32. Two coils have self inductances of 250 mH and 400 mH respectively. Determine the magnetic coupling coefficient of the pair of coils if their mutual inductance is 80 mH.

A. 0.115 C. 0.124
B. 0.253 D. 0.162

33. A chip resistor is marked 394. Its resistance value is

A. 39.4 Ω
B. 390,000 Ω
C. 39,000 Ω
D. 394 Ω

34. The best place to use a wirewound resistor is ______.

A. in a radio-frequency amplifier
B. when the resistor doesn't dissipate much power
C. in a high-power, radio frequency circuit
D. in a high-power, direct-current circuit

35. Self-inductance occurs when the:

A. current is changing
B. circuit is changing
C. flux is changing
D. resistance is changing

36. A capacitor uses a dielectric 0.04 mm thick and operates at 30 V. What is the electric field strength across the dielectric at this voltage?

A. 750 kV/m
B. 650 kV/m
C. 850 kV/m
D. 700 kV/m

37. Calculate the equivalent capacitance of two capacitors of 6 μF and 4 μF connected in parallel and in series.

A. 10 μF; 2.4 μF
B. 10 mF; 2.4 mF
C. 6 μF; 1.5 μF
D. 6 mF; 1.5 mF

38. Which of the following resistors has the smallest physical size?

A. wire-wound resistors.
B. carbon-composition resistors.
C. surface-mount resistors.
D. potentiometers.

39. Which of the following statement is false?

A. An air capacitor is normally a variable type
B. A paper capacitor generally has a shorter service life than most other types of capacitor
C. An electrolytic capacitor must be used only on a.c. supplies
D. Plastic capacitors generally operate satisfactorily under conditions of high temperature

40. If a thermistor has a negative temperature coefficient (NTC), its resistance

A. increases with an increase in operating temperature.
B. decreases with a decrease in operating temperature.
C. decreases with an increase in operating temperature.
D. is unaffected by its operating temperature.

41. Two coils connected in series have self inductance of 40 mH and 10 mH respectively. The total inductance of the circuit is found to be 60 mH. Determine the mutual inductance between the two coils and the coefficient of coupling.

A. 5 H, 0.23
B. 5 μH, 0.21
C. 5 mH, 0.25
D. 5 nH, 0.24

42. The capacitance of a capacitor is the ratio

A. charge to p.d. between plates
B. p.d. between plates to plate spacing
C. p.d. between plates to thickness of dielectric
D. p.d. between plates to charge

43. The effect of inductance occurs in an electrical circuit when:

A. the resistance is changing
B. the flux is changing
C. the current is changing
D. all of the above

44. A current changing at a rate of 5 A/s in a coil of inductance 5 H induces an e.m.f. of:

A. 25 V in the same direction as the applied voltage
B. 1 V in the same direction as the applied voltage
C. 25 V in the opposite direction to the applied voltage
D. 1 V in the opposite direction to applied voltage

45. Which of the following axial-lead resistor types usually has a blue, light green or red body?

A. wire-wound resistors.
B. carbon-composition resistors.
C. carbon-film resistors.
D. metal-film resistors.

46. The resistance of 500 m of wire of cross-sectional area 2.6 mm^2 is 5 Ω. Determine the resistivity of the wire in μΩ-m.

A. 0.26 μΩ-m
B. 0.026 μΩ-m
C. 2.60 μΩ-m
D. 0.0026 μΩ-m

47. If a 75-ohm sample of copper wire at 0°C is heated to 30°C, what is the approximate total resistance? (The temperature coefficient of resistance of copper at 0°C is 0.00427).

A. 0.32 Ω
B. 9.61 Ω
C. 65.39 Ω
D. 84.61 Ω

48. Which of the following statements is true?

A. Resistors always have axial leads.
B. Resistors are always made from carbon.
C. There is no correlation between the physical size of a resistor and its resistance value.
D. The shell life of a resistor is about 1 year.

49. Two metal plates separated by air form a 0.001 μF capacitor. Its value may be changed to 0.002 μF by:

A. bringing the metal plates closer together
B. using the plates smaller in size
C. moving the plates apart
D. touching the two plates together

50. Find the charge on a 10 μF capacitor when the applied voltage is 250 V.

A. 2.5 mC
B. 1.5 mC
C. 3.0 mC
D. 5.0 mC

END